

Ahmed Abdelrazek

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SUMMARY

AI Engineer specializing in **enterprise AI systems**, **Generative AI**, and **production-ready machine learning workflows**. Experienced in developing scalable **LLM-powered applications**, **Retrieval-Augmented Generation (RAG) pipelines**, and **digital twin AI systems** within multi-tenant SaaS architectures. Strong background in **Python**, **PyTorch**, **TensorFlow**, **FastAPI**, **Docker**, and cloud-based ML workflows (AWS & Azure exposure). Passionate about building reliable, scalable AI systems aligned with real-world business impact.

EDUCATION

Egypt-Japan University of Science and Technology (E-JUST)

B.Sc. in Artificial Intelligence & Data Science

Alexandria, Egypt

2021 – 2025

EXPERIENCE

AI Engineer

Jul 2025 – Mar 2026

Performance Leap – Enterprise AI Digital Twins (Remote, USA)

- Developing AI models powering enterprise-grade digital twins for intelligent meeting optimization.
- Contributing to personalization pipelines modeling user communication styles, decision patterns, and expertise.
- Working within a scalable multi-tenant SaaS architecture designed for enterprise environments.
- Supporting integration of AI assistants with Zoom, Microsoft Teams, and Google Meet platforms.
- Participating in research-driven AI development including adaptive modeling and real-time inference optimization.
- Collaborating remotely with cross-functional research and engineering teams.

Instructor

Feb 2025 – Sep 2025

iSchool (Part-time, Remote)

Egypt

- Delivered technical instruction in programming and AI fundamentals to students in structured learning programs.
- Mentored students through hands-on projects and problem-solving sessions.

Software Developer (Internship)

Jun 2025 – Aug 2025

International Group For Modern Coatings (MIDO)

Alexandria, Egypt

- Developed software solutions and contributed to backend systems in a production environment.

AI Specialist

Dec 2024 – May 2025

Outlier (Contract)

Remote

- Contributed to development and evaluation of large-scale Generative AI systems.
- Optimized prompt engineering strategies and ML workflows to improve response reliability.
- Conducted structured experimentation to enhance model robustness and scalability.
- Delivered production-quality Python code for next-generation AI applications.

Freelance Data Science & AI Engineer

Oct 2024 – Mar 2025

Self-Employed

Remote

- Designed and implemented end-to-end ML pipelines (data ingestion, preprocessing, training, validation, inference).
- Built RESTful inference APIs using FastAPI for scalable AI model serving.
- Containerized ML services using Docker for reproducible deployments.
- Worked in cloud-based ML environments (AWS & Azure exposure) for training and deployment workflows.

Software Developer (Internship)

Jun 2025 – Sep 2025

Emaar Misr – Marassi

Egypt

- Developed backend services and APIs following modern software engineering practices.
- Improved system reliability through structured logging and data validation.

AI Intern – Microsoft Machine Learning Engineer Track

May 2024 – Oct 2024

Digital Egypt Pioneers Initiative (DEPI)

Egypt

- Built and optimized machine learning, deep learning, and NLP models for applied AI use cases.
- Awarded **1st Place** for best AI project across the Microsoft ML Engineer track.

AI Intern – Huawei Artificial Intelligence Track

Sep 2024

Huawei

Egypt

- Developed deep learning models for computer vision and AI system pipelines.

PROJECTS

AI Multi-Agent Product Integration Hub | *Python, FastAPI, LangGraph, PostgreSQL, Redis, Docker*

- Designed a production-oriented multi-agent LLM system with structured tool orchestration.
- Built scalable API endpoints and persistent storage mechanisms for reliable system behavior.
- Implemented retry logic and queue-based processing for improved fault tolerance.

University AI Assistant Agent (RAG System) | *Python, LangChain, ChromaDB, Flask*

- Developed Retrieval-Augmented Generation (RAG) pipeline using embeddings and semantic search.
- Applied document chunking strategies and role-based access control (RBAC).
- Integrated live querying to improve contextual accuracy of AI responses.

Real-Time ASL GPT | *Python, PyTorch, MediaPipe, Neptune*

- Processed 200GB+ dataset and trained SqueezeFormer-based model achieving 89% validation similarity.
- Developed low-latency inference pipeline for real-time gesture recognition.
- Tracked experiments and model performance using Neptune.

Medical X-ray Image Classification | *Python, TensorFlow*

- Built CNN-based classifier for pneumonia detection with data augmentation and hyperparameter tuning.

AWARDS & CERTIFICATIONS

1st Place – 3rd International AI Forum

Feb 2025

Ministry of Communications and Information Technology (MCIT), Egypt

- Awarded first place among 430+ AI projects for an innovative artificial intelligence solution.

1st Place – DEPI AI & Data Science Track (Microsoft ML Engineer)

Nov 2024

Digital Egypt Pioneers Initiative (DEPI)

- Ranked first across the Microsoft Machine Learning Engineer track for best-performing AI project.

Microsoft Machine Learning Engineer Certification

2025

Digital Egypt Pioneers Initiative (DEPI)

- Certified in machine learning engineering, NLP, deep learning, and production-ready AI systems.

Huawei Certified AI Associate (HCIA-AI v3.5)

2024

Huawei

- Certification covering enterprise AI platforms, deep learning, computer vision, and AI system design.

Artificial Intelligence Certification

2024

National Telecommunication Institute (NTI)

- Hands-on certification in machine learning, NLP, data analysis, and big data fundamentals.

Oracle Database SQL Certified Associate

2024

Oracle

- Validated skills in SQL, relational database design, and data querying for production systems.

COURSES

Machine Learning with Python: From Linear Models to Deep Learning (MITx 6.86x)

2024

Massachusetts Institute of Technology (edX)

- Covered linear models, neural networks, CNNs, and deep learning optimization techniques.

Probability: The Science of Uncertainty and Data (MITx 6.431x)

2024

Massachusetts Institute of Technology (edX)

- Studied probability theory, Bayesian inference, and statistical reasoning for ML systems.

AI & Data Science – Microsoft Machine Learning Engineer Track

2024

Digital Egypt Pioneers Initiative (DEPI)

- Comprehensive training in ML, NLP, deep learning, and production AI workflows.

Huawei Certified AI Associate (HCIA-AI v3.5)

2024

Huawei

- Covered enterprise AI platforms, deep learning, and computer vision systems.

Cloud and Virtualization Concepts

2024

ITI – MaharaTech

- Studied cloud infrastructure and virtualization supporting scalable ML systems.

TECHNICAL SKILLS

Programming: Python, SQL, TypeScript, JavaScript

Machine Learning: PyTorch, TensorFlow, NLP, Computer Vision, Statistical Modeling

Generative AI: LLMs, RAG, Prompt Engineering, Embeddings, LangChain, LangGraph

Cloud: AWS (ML workflow exposure), Azure (ML workflow exposure), Cloud Infrastructure Concepts

Deployment & DevOps: Docker, CI/CD, FastAPI, REST APIs, Model Serving, Experiment Tracking

Databases: PostgreSQL, Redis

Tools: Git, OpenTelemetry

LANGUAGES

Arabic (Native), English (Professional)